



Experiment 9

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CLASS: SE-2

Title

Implement search functionality to filter API results dynamically.

Theory:

Modern web applications often handle large amounts of data retrieved from external APIs. Displaying all fetched data at once can reduce usability. To improve user experience, search and filtering functionality is implemented to dynamically narrow down results based on user input.

An API (Application Programming Interface) provides structured data in formats such as JSON. JavaScript fetches this data asynchronously and stores it in memory. A search input field allows users to enter keywords, which are used to filter the API results in real time.

1. Dynamic Search Filtering

Dynamic search filtering allows data to be filtered without refreshing the page. As the user types in the search box, JavaScript compares the input text with the fetched data and displays only matching records.

2. AJAX and Fetch API

The Fetch API is used to retrieve data asynchronously from the server. It returns a promise that resolves with the API response, which is then converted into JSON format. This ensures smooth and fast data retrieval.

3. JavaScript Array Methods

JavaScript provides built-in array methods such as `filter()` and `includes()` to efficiently search through data. These methods help compare user input with data fields like name, title, or description.

4. DOM Manipulation

The filtered data is displayed dynamically using DOM manipulation, ensuring instant updates to the user interface whenever the search input changes.



Algorithm / Procedure

1. Start the experiment.
2. Create an HTML file with a search input field.
3. Use JavaScript to fetch data from a public API.
4. Store the API response in an array.
5. Display all API results on the webpage.
6. Capture user input from the search field.
7. Filter the stored data based on the search keyword.
8. Update the displayed results dynamically.
9. Repeat filtering as the user types.
10. Stop the experiment.

Program / Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>API Search Filter</title>

<style>    body {        font-
family: Arial;        background:
#f4f4f4;
```



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```
        }      input {      width:
300px;      padding: 10px;
margin: 20px;      }

        .card {
background: white;
padding: 15px;
margin: 10px;      border-
radius: 5px;
        }
</style>
</head>
<body>

<h2 align="center">Search API Data</h2>
<center>
<input type="text" id="search" placeholder="Search by title...">
</center>

<div id="results"></div>

<script> let
apiData = [];

fetch("https://jsonplaceholder.typicode.com/posts")
.then(response => response.json())
```



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```
.then(data => {  
  apiData = data;  
  displayData(apiData);  
});
```

```
function displayData(data) {  const resultDiv =  
  document.getElementById("results");  resultDiv.innerHTML  
  = "";
```

```
    data.forEach(item => {      resultDiv.innerHTML  
+= `  
<div class="card">  
    <h4>${item.title}</h4>  
    <p>${item.body}</p>  
  </div>  
  `;  
    });  
}
```

```
document.getElementById("search").addEventListener("keyup", function() {  
  const keyword = this.value.toLowerCase();  const filteredData =  
  apiData.filter(item =>    item.title.toLowerCase().includes(keyword)  
  );  
  displayData(filteredData);  
});  
</script>
```



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</body>

</html>

Scenario based questions:

Create a web page that fetches data from an API and allows users to search and filter the displayed results dynamically using a search input box.

Output:

User Search (API Data)

Search by name...

Name: Leanne Graham
Email: Sincere@april.biz
City: Gwenborough

Name: Ervin Howell
Email: Shanna@melissa.tv
City: Wisokyburgh

Name: Clementine Bauch
Email: Nathan@yesenia.net
City: McKenziehaven

Name: Patricia Lebsack
Email: Julianne.OConner@kory.org
City: South Elvis

Name: Chelsey Dietrich
Email: Lucio_Hettinger@annie.ca
City: Roscoeview

Name: Mrs. Dennis Schulist
Email: Karley_Dach@jasper.info
City: South Christy

Name: Kurtis Weissnat
Email: Telly.Hoeger@billy.biz
City: Howemouth

Conclusion

Thus, dynamic search functionality was successfully implemented to filter API results using JavaScript. This experiment demonstrates the use of Fetch API, array filtering methods, and DOM manipulation to create responsive and interactive web applications.



Question:

1. What is dynamic search filtering?

Dynamic search filtering is a UI pattern where a list of items is filtered in real-time based on user input. Instead of typing a query and hitting a "Submit" button, the application listens for every keystroke and immediately hides items that don't match the search criteria.

2. How is DOM manipulation used in this experiment?

In a search experiment, JavaScript uses DOM (Document Object Model) manipulation to change what the user sees without fetching a new page. It usually works like this:

- **Capturing Input:** JavaScript "grabs" the value from the search input field using event listeners (like input or keyup).
- **Traversing the List:** It loops through the existing HTML elements (like or <div> tags) that represent your data.
- **Modifying Styles:** For each element, JavaScript checks if the text matches the search term.
 - If it **matches**, it ensures the element is visible.
 - If it **doesn't match**, it manipulates the DOM to set the element's style to display: none.

3. Why is page reload not required?

A page reload is unnecessary because the data is **already present** in the browser's memory or the HTML structure.

- **Client-Side Processing:** Since JavaScript is running directly in the user's browser, it can modify the existing HTML structure (the DOM) instantly.
- **No Server Trip:** A page reload happens when the browser asks the server for a whole new document. In dynamic filtering, we aren't asking for new info; we are just choosing which bits of the *current* info to show or hide.
- **State Management:** By avoiding a reload, the user doesn't lose their place, the "focus" remains in the search bar, and the experience feels seamless.



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